JEMAY SALOMON

Ph.D. Candidate in Quantitative Genetics

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SUMMARY

I am a PhD student in quantitative genetics. I'm interested in using advanced mixed-modeling approaches to better estimate genetic effects and (co)variance components (e.g. automatic differentiation using R/C++), evaluating model robustness through stochastic simulations, and, in a more general way, applying quantitative genetics theory to dissect the genetic architecture of complex traits.

EDUCATION

Ph.D. in Quantitative Genetics
Université Paris-Saclay, Gif-sur-Yvette, FR
Oct. 2023 – Present

M.Sc. in Quantitative Genetics
Institut Agro Montpellier, Montpellier, FR
Sept. 2021 – Sept. 2023

B.Sc. in Agronomy
Université Quisqueya, Port-au-Prince, HT
Sept. 2013 – Sept. 2018

PROFESSIONAL EXPERIENCE

Research Assistant | INRAE-GQE_Le Moulon, Gif-sur-Yvette, FR
Oct. 2023 – Present

- Investigate the genetic basis and architecture of interspecific mixing ability in wheat-pea association
- Develop a joint breeding methodology for sole and intercropping systems
- Design and implement research protocol

Research Intern | CHIBAS, Mirebalais, HT
Mar. 2023 – Aug. 2023

- Optimized genomic prediction models for sorghum yield across multiple environments
- Designed and implementing research protocol
- Collected and managed phenotypic data from field trials, contributing to high-quality dataset creation

Research Intern | CIRAD-CNRS, Montpellier, FR
May 2022 – Aug. 2022

- Modeled phyllochron-plastochron relationship in sorghum in order to predict panicle initiation date
- Supported field trial operations through phenotypic data collection
- Synthesized findings into a scientific report, presenting analytical results and growth model interpretations

PUBLICATIONS

Preprints

Salomon, J., Enjalbert, J., & Flutre, T. (2026). Joint modeling of social genetic effects in mono- and pluri-specific groups : Case study in intercrops. *bioRxiv*, 2026.03.27.714849. https://doi.org/10.64898/2026.03.27.714849

SOFTWARE

Flutre, T., INRAE, Enjalbert, J., Forst, E., Remerand, M., & **Salomon, J**. (2026). plantmix : Genetic Study of Plant Mixtures (Version 1.0.2) [Logiciel]. https://cran.r-project.org/web/packages/plantmix/index.html

CONFERENCE PRESENTATIONS

Salomon, J., Enjalbert, J., & Flutre, T. (2025, September 17). Leveraging a broad gradient of plant–plant interactions to efficiently breed for cereal–legume mixtures. Poster presented at the XIX Conference of EUCARPIA Biometrics in Plant Breeding. https://hal.inrae.fr/hal-05546246

Salomon, J., Enjalbert, J., & Flutre, T. (2025, March 10-14). Leveraging genetic diversity to study plant-plant interactions and breed for crop mixtures. Poster presented at the PSL-Qlife Quantitative Biology Winter School 2025.

Salomon, J., Enjalbert, J., & Flutre, T. (2024, October 22). Leveraging a broad gradient of plant–plant interactions to efficiently breed for cereal–legume mixtures. Oral presentation at SFE2 Conference.

Salomon, J., & Pressoir, G. (2023). Optimization of prediction models for yield stability in sorghum: Analysis of physiological determinisms, genomic prediction incorporating G×E, and phenomic approaches. Poster presented at ELLS Scientific Student Conference 2023.

MENTORING

Rouguiyatou Diallo (2025/04 – 2025/8: Co-Supervision with Timothée Flutre): Master1's thesis topic: *Characterize the genetic basis of phenotypic plasticity in wheat grown in solecrop versus in intercrop with pea, in terms of growth and epidemic limitation.*

Viéban Rémi (2026/06/01 – 2026/06/19): Internship topic (Engineer 1st year): *Evaluation of phenomic and genomic prediction in cereal-legume intercropping systems.*

INTERESTS

- Music Production: Managing a private record label
- DJ & Live performance

SKILLS

- Quantitative Genetics
- Plant Breeding
- Intercrop Breeding
- Genomic Selection
- Statistical Genetics
- R Programming
- C++ Programming (intermediate)
- Linear Mixed Models
- Statistical Modeling

REFEREES

- **Dr. Jerôme Enjalbert,** Research Director, Génétique Quantitative et Évolution - Le Moulon, Université Paris-Saclay, INRAE, CNRS, AgroParisTech. Email: jerome.enjalbert@inrae.fr; Tel: 01 69 15 68 55
- **Dr. Timothée Flutre,** Research Scientist, Génétique Quantitative et Évolution - Le Moulon, Université Paris-Saclay, INRAE, CNRS, AgroParisTech. Email: timothee.flutre@inrae.fr; Tel: 01 69 15 61 33
- **Dr. Dominique This,** Professor, Institut Agro - Montpellier, UMR AGAP Institut. Email: dominique.this@supagro.fr ; Tel: 4 67 61 58 29 | 6 20 80 63 39